

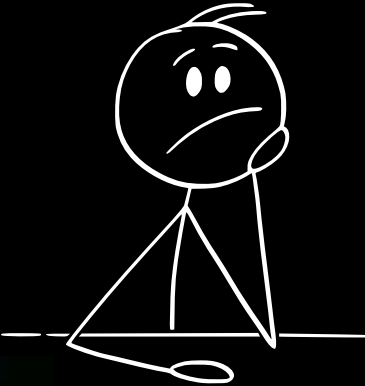


TEAM NAME: PHARMA INNOVATORS

COLLEGE NAME: SRI VENKATESWARA COLLEGE OF ENGINEERING

PROBLEM STATEMENT: THE DRUG REPURPOSING DISCOVERY
CHALLENGE





PROBLEM STATEMENT

The Business Context

A leading multi-nation generic pharmaceutical company with significant US business seeks to diversify beyond the highly competitive, low-margin generics market by developing **value-added innovative products** through drug repurposing.

The Core Challenge

Identifying opportunities for repurposing approved molecules requires extensive manual literature reviews across disconnected sources, taking **2-3 months** with multiple iterations to uncover viable product concepts.

Regulatory
FDA, WHO

Clinical Trials
CT.gov

Scientific
PubMed

Commercial
IQVIA, EXIM



 **2-3 MONTHS MANUAL
RESEARCH & SYNTHESIS**



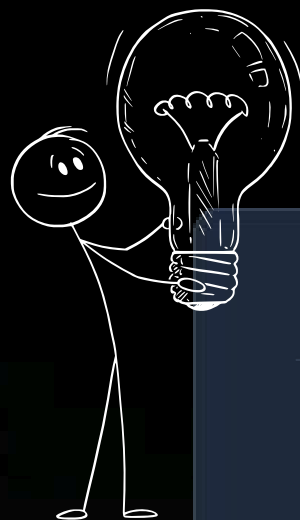
 **Missed Opportunities & Slow Pipeline Decisions**

The Goal

Design an **Agentic AI solution** where a Master Agent:

- ✓ Integrates with regulatory websites, clinical trials, scientific journals, paid databases & internal repositories.
- ✓ Allows users to input prompts for web search, market analysis & journal summarization.
- ✓ Generates summary reports and saves to archival system.

Target: Reduce research time from **months to minutes** while increasing throughput of quality evaluations.



SOLUTION



💡 Solution Scenario

- Agentic AI platform for interactive molecule evaluation via user prompts
- Master Agent orchestration that decomposes queries and routes tasks to Worker Agents
- Multi-source intelligence retrieval (market, clinical trials, patents, literature, trade, internal data)
- Synthesized innovation assessment exported as a structured report

🏢 Target Audience & Industry

Target Industry: R&D / PortfolioManagement / Competitive Intelligence

Industry Type: B2B

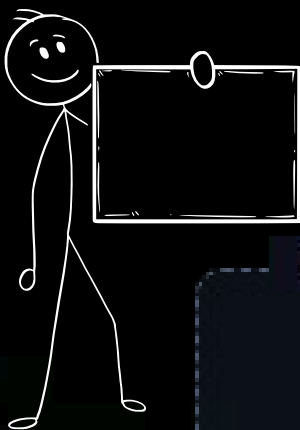
User Group: PortfolioStrategy & Business Development Teams

User Department: R&D / PortfolioManagement / Competitive Intelligence

≡ Proposed Data Flow & Output

User prompt → Master Agent → Worker Agents (IQVIA, EXIM, Patent, CT.gov, Web, Internal) → Aggregation by Master Agent → Report Generator → PDF archived+ delivered.

Nature of Output: Web application with downloadable PDF reports.



FLOW DIAGRAM



USER INTERFACE LAYER

React.js Web Application

- Chat Interface for Natural Language Queries
- Real-time Research Status Dashboard



ORCHESTRATION LAYER

LangGraph Workflow Engine

- Manages State & Task Execution Graph
- Coordinates Worker Agent Sequence



REASONING CORE

Fine-tuned DeepSeek model

Master Agent • Query Analysis • Synthesis

WORKER AGENTS LAYER

IQVIA

Patent

Clinical

EXIM

Web

Report

Internal

DATA & STORAGE LAYER

MongoDB

User Sessions • ChatHistory • Audit

Vector DB

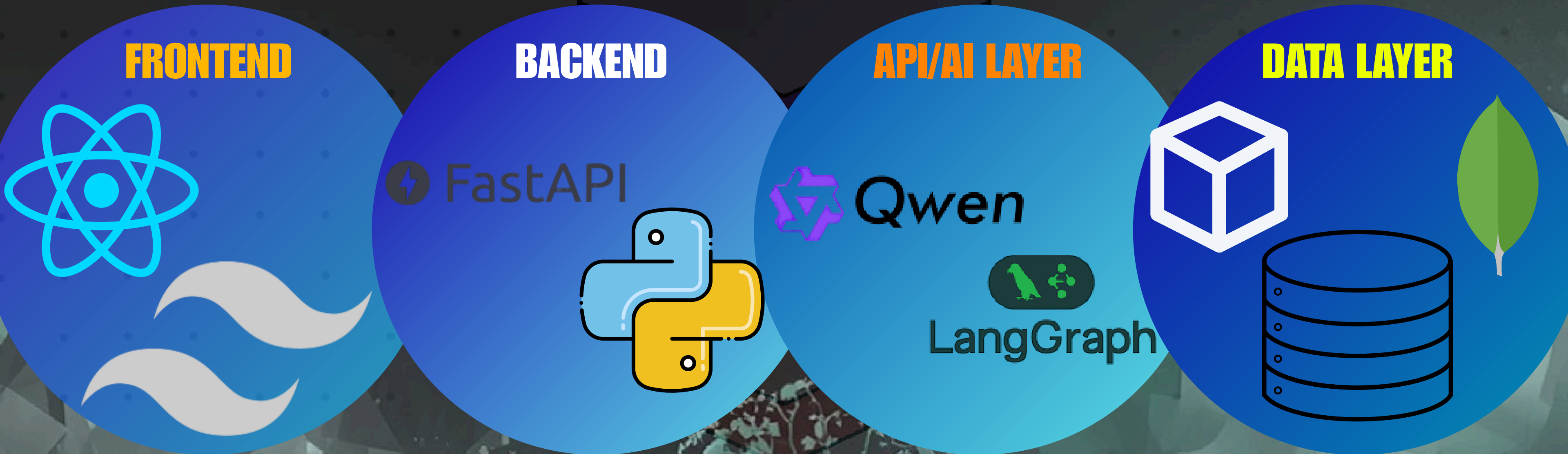
FDA Docs • Scientific Papers

External APIs

ClinicalTrials.gov • USPTO • IQVIA



TECHNICAL REQUIREMENTS



USECASE / NOVELTY

★ Key Novelty Factors

Novelty Element	Description
LangGraph Stateful Orchestration	Persistent conversation memory across multi-turn research with conditional branching
Fine-tuned Qwen Master Agent	Domain-adapted on pharma literature for medical terminology accuracy
Multi-Agent Parallel Execution	8 specialized agents working concurrently (60x faster than manual)
Vector RAG Pipeline	Pinecone semantic search on FDA docs, scientific papers & internal documents
End-to-End Automation	Natural language query → parallel research → PDF report in <30 seconds

Representative Use Cases

Use Case	Prompt Example	Business Value
Market Gap Analysis	"Respiratory diseases with low competition in India?"	Identify white spaces
Repurposing Scan	"Oncology molecules for autoimmune indications"	Extend asset lifecycle
Patent Intelligence	"Patent expiry for top 10 cardiovascular drugs"	Plan generic entry
Clinical Landscape	"Phase 3 trials for GLP-1 beyond diabetes"	Track competitors
Portfolio Optimization	"Our expiring assets suitable for reformulation"	Maximize ROI

🏆 Competitive Advantage

Parameter	Manual Process	Our Solution
Research Time	2-3 months	<30 minutes
Sources Checked	5-10	50+ automated
Report Generation	3-5 days	30 seconds
Cost per Analysis	\$15-25K	\$500-2K



BUSINESS MODEL / TARGET MARKET



REVENUE CHANNELS

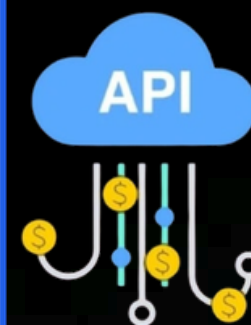
SaaS Subscriptions –
Recurring monthly
or annual fees for
platform access

API Licensing –
Usage-based billing
for integrating data
into external apps

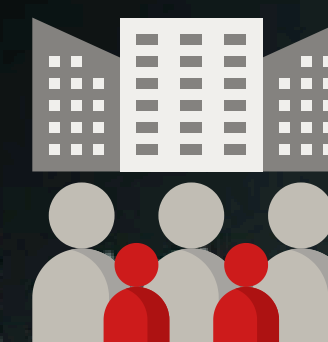
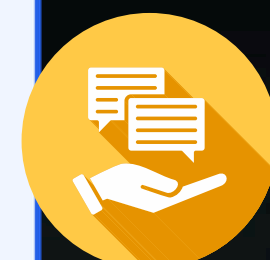
**Enterprise
Customization –**
High-ticket bespoke
features and private
cloud hosting

**Professional
Services – Expert**
data consulting and
specialized research
reports

**Maintenance
Contracts –**
Guaranteed uptime
and dedicated
technical support



Delivers Value To



MARKET SEGMENTS

Big Pharma – Global
corporations
requiring large scale
drug discovery tools

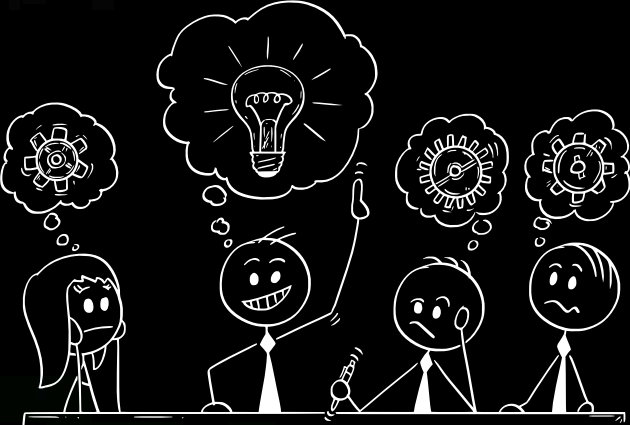
Biotech Startups –
Agile teams focused
on specific
therapeutic
innovations

**Academic
Institutions –**
Universities
conducting
fundamental
biological research

CROs –
Organizations
managing
outsourced clinical
trials and data

**Public Health
Agencies –**
Government bodies
tracking health
trends and efficacy





TEAM DETAILS



ROLE	NAME	STREAM/DEPT	YEAR	COLLEGE
Fullstack/Agentic AI	Tanish M	BE/CSE	3rd year	SVCE
Fullstack/Agentic AI	Yuvaranjan S	BE/CSE	3rd year	SVCE
Fullstack	Tarun Kumar R	BE/CSE	3rd year	SVCE
Fullstack	Aswin S	BE/IT	3rd year	SVCE